



Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY OF TECHNOLOGY

Data Analytics and Data Visualization Laboratory
(303105315)

6th SEMESTER

Laboratory Manual

CERTIFICATE

This is to certify that ..DHRUV...TIWARI... with enrolment no.
230303154036 and 6th Semester...GIEP-2 has successfully
completed her laboratory experiments in the Data Analytics and Data Visualization
(303105315) from the department of Computer Science & Engineering during the academic
year ..2023-24..2024.



Date of Submission:

Staff In charge: 

Head Of Department:

Preface

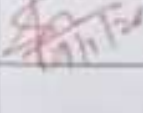
It gives us immense pleasure to present the first edition of the Data Analytics and Data Visualization Practical Book for the B. Tech. 5th Semester students of PARUL UNIVERSITY. The Data Analytics and Visualization laboratory courses at PARUL UNIVERSITY, WAGHODIA, are designed to help students bridge the gap between theoretical knowledge and real-world application. Students are introduced to fundamental concepts in the classroom and then gain hands-on experience through laboratory exercises involving data analysis, preprocessing, modeling, and visualization techniques.

The main objective of the Data Analytics and Data Visualization Practical Book is to provide practical exposure to students in the domain of data science by enabling them to explore datasets, apply statistical and machine learning methods, and visualize insights for informed decision-making. The experiments are structured to offer clarity on how to handle data pipelines, choose appropriate analytical models, and create meaningful visual representations using tools like Python, R, Power BI, or Tableau.

This Practical Book aims to be a comprehensive guide for all experiments included in the Data Analytics and Data Visualization laboratory course. It covers various aspects of each experiment, such as understanding the data structure, applying suitable algorithms, interpreting model performance, and constructing informative and interactive visuals. It also includes detailed instructions on data handling, result computation, and discussions on findings and insights derived from the analysis.

We sincerely acknowledge the authors and publishers of the reference materials that guided us during the preparation of this Practical Book. We hope this Data Analytics and Data Visualization Practical Book proves to be a valuable resource for students and helps them build a strong foundation for advanced studies and professional practice in the field of data science.

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Enrollment No: 2303031540130

Practical-1

Aim: Perform Exploratory Data Analysis on the given dataset using Python.

Procedure:

1. Import the dataset
2. View the head of the data
3. View the basic information of data and description of data
4. Find the unique value of data and verify the duplication of data
5. Plot a graph for unique value of dataset
6. Verify the presence of null value and replace the null value
7. Visualize the needed data

Program:

```
#Load the required libraries
import pandas as pd
import numpy as np

import seaborn as sns
#Load the data

df = pd.read_csv('titanic.csv')

#View the data
df.head()

df.info()
df.describe()
#Find the duplicates
df.duplicated().sum()

#unique values
df['Pclass'].unique()
df['Survived'].unique()
df['Sex'].unique()
    array([3, 1, 2], dtype=int64)
    array([0, 1], dtype=int64)
    array(['male', 'female'], dtype=object)

#Plot the unique values
sns.countplot(df['Pclass']).unique()

#Find null values
df.isnull().sum()

#Replace null values
df.replace(np.nan,'0',inplace = True)
```

Enrollment No:

```
#Check the changes now  
df.isnull().sum()
```

```
#Filter data  
df[df['Pclass']==1].head()
```

```
#Boxplot  
df[['Fare']].boxplot()
```

Practical-2

Aim: Calculate mean, median and mode of the first 50 records in the given dataset using python

```
from statistics import median

import pandas as pd

import matplotlib.pyplot as plt

d = pd.read_excel(r'F:\SEMESTER 5\DATA VISULIZATION AND DATA
ANALYTICS\LAB\Marks.xlsx')

f = pd.DataFrame(d)

print(f)

meanm = d["Maths"].mean()

medianm = d["Maths"].median()

modem = d["Maths"].mode()

meanp = d["Physics"].mean()

medianp =

d["Physics"].median()modep =

d["Physics"].mode() cor =

d.corr()

print("Maths:")

print("Mean: ",meanm)

print("Median: ",medianm)

print("Mode: ",modem)

print("Physics:")

print("Mean: ",meanp)

print("Median: ",medianp)

print("Mode: ",modep)
```

Enrollment No:

```
print("Correlation: ")

print(cor)

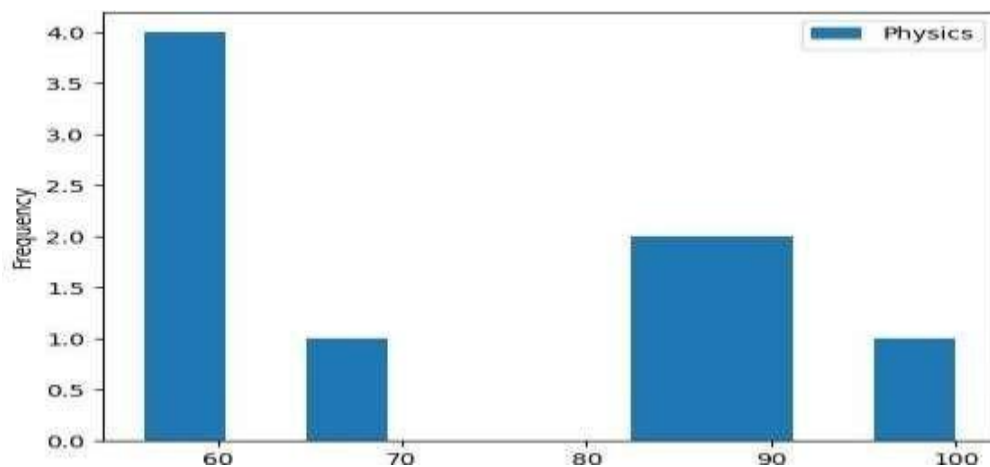
d.plot(kind = 'hist', x = 'Maths', y =
'Physics')plt.show()
```

OUTPUT:

```
S.No.  Maths  Physics
0      1     90      59
1      2     61      57
2      3     90      56
3      4     75      89
4      5     53      83
5      6     53      89
6      7     73      85
7      8     89     100
8      9     79      56
9     10     64      69

Maths:
Mean:   72.7
Median:  74.0
Mode:    0    53
1     90
dtype: int64
Physics:
Mean:   74.3
Median:  76.0
Mode:    0    56
1     89
dtype: int64
Correlation:

      S.No.  Maths  Physics
S.No.  1.000000 -0.150359  0.326068
Maths  -0.150359  1.000000 -0.207416
Physics 0.326068 -0.207416  1.000000
```



Enrollment No:

Practical-3

Aim: Perform Multiple Linear Regression on data.

```
import pandas as pd

import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import r2_score

import matplotlib.pyplot as plt

df=pd.read_excel(r'H:\CODES\PYTHON\JEET.xlsx')

df.head()

x=df.drop(['V'],axis=1).value

sy=df['V'].values

print(x)

print(y)

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=0)

ml=LinearRegression()

ml.fit(x_train,y_train)

y_pred=ml.predict(x_test)

print(y_pred)

r2_score(y_test,y_pred)

plt.figure(figsize=(15,10))

plt.scatter(y_test,y_pred)

plt.xlabel('Original')
```

Enrollment No:

```
plt.ylabel('Predicted')

plt.title('Original vs Predicted')

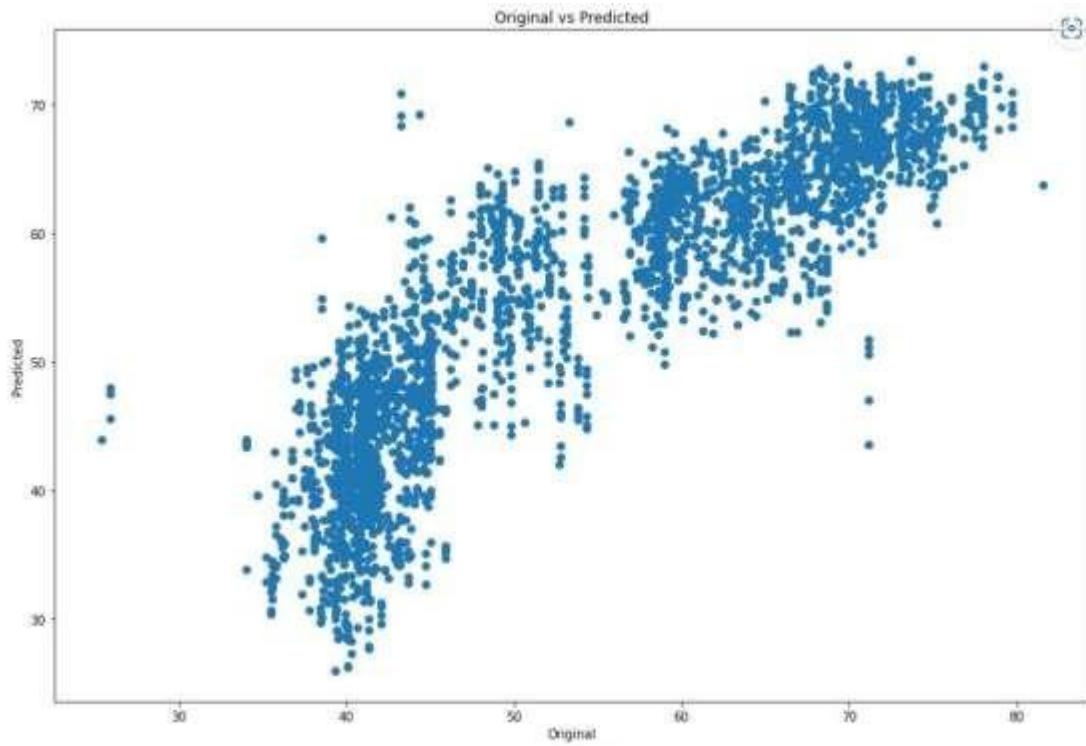
pred_y_df=pd.DataFrame({'Original Value':y_test,'Predicted
Value':y_pred,'Difference':y_test-y_pred})

pred_y_df[0:20]
```

OUTPUT

```
[[ 14.96 1024.07  73.17  463.26]
 [ 25.18 1020.04  59.08  444.37]
 [  5.11 1012.16  92.14  488.56]
 ...
 [ 31.32 1012.92  36.48  429.57]
 [ 24.48 1013.86  62.39  435.74]
 [ 21.6  1017.23  67.87  453.28]]
[41.76 62.96 39.4 ... 74.33 69.45 62.52]
[69.43492077 53.10902176 51.95452875 ... 71.36215109 67.45722613
 63.9412364 ]
```

	Original Value	Predicted Value	Difference
0	77.95	69.434921	8.515079
1	49.39	53.109022	-3.719022
2	43.14	51.954529	-8.814529
3	58.33	59.418508	-1.088508
4	55.28	55.427943	-0.147943
5	74.87	68.948525	5.921475
6	43.56	38.994669	4.565331
7	50.59	52.288060	-1.698060
8	41.54	39.753457	1.786543
9	50.16	59.930829	-9.770829
10	63.73	61.956249	1.773751
11	69.45	64.296091	5.153909
12	58.79	62.413643	-3.623643
13	63.21	64.865435	-1.655435
14	74.22	65.665335	8.554665
15	45.38	53.122579	-7.742579
16	64.63	59.613898	5.016102
17	68.12	66.702570	1.417430
18	58.49	56.387461	2.102539
19	73.42	70.605629	2.814371



Practical-4

Aim: Perform the Logistic Regression on a dataset.

LOGISTIC REGRESSION: Logistic regression estimates the probability of an event occurring, such as voted or didn't vote, based on a given dataset of independent variables

CODE:

```
from sklearn.datasets import load_digits

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

digits = load_digits()

dir(digits)

digits.data[4]

plt.gray()

plt.matshow(digits.images[1])

digits.target[0:5]

x_train,x_test,y_train,y_test = train_test_split(digits.data,digits.target,test_size=0.2)

len(x_test)

logistic = LogisticRegression()

logistic.fit(x_train,y_train)

logistic.score(x_test,y_test)

plt.matshow(digits.images[84])

digits.target[84]
```

Enrollment No:

```
logistic.predict([digits.data[84]])
```

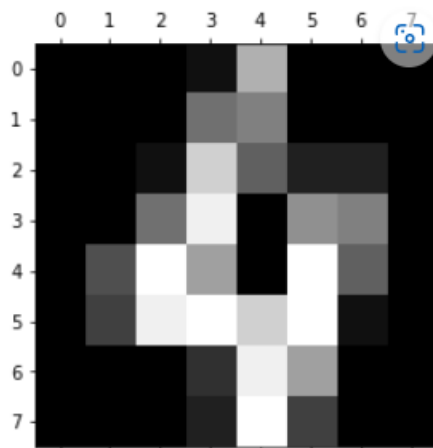
OUTPUT

```
['DESCR', 'data', 'feature_names', 'frame', 'images', 'target', 'target_names']
```

```
array([ 0.,  0.,  0.,  1., 11.,  0.,  0.,  0.,  0.,  0.,  7.,  8.,
        0.,  0.,  0.,  0.,  0.,  1., 13.,  6.,  2.,  2.,  0.,  0.,  0.,
        7., 15.,  0.,  9.,  8.,  0.,  0.,  5., 16., 10.,  0., 16.,  6.,
        0.,  0.,  4., 15., 16., 13., 16.,  1.,  0.,  0.,  0.,  0.,  3.,
       15., 10.,  0.,  0.,  0.,  0.,  0.,  2., 16.,  4.,  0.,  0.]])
```

```
: <matplotlib.image.AxesImage at 0x24a08d62e50>
```

```
<Figure size 432x288 with 0 Axes>
```

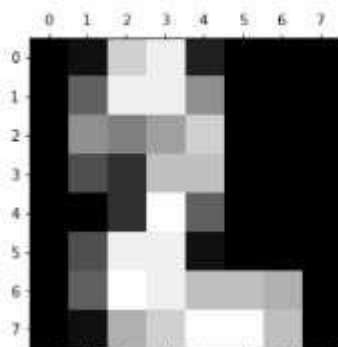


```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
360
```

```
0.9444444444444444
```

```
<matplotlib.image.AxesImage at 0x24a08b2b5b0>
```



```
array([2])
```

Practical-5

Aim: Use a dataset & apply K means clustering to get insights from data.

K MEANS CLUSTING: K means clustering is one of the simplest algorithm which uses unsupervised learning method to solve known clustering issues

CODE:

```
In [7]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

```
In [5]: df = pd.read_csv(r'H:\SEMESTER 5\DATA VISULIZATION AND DATA ANALYTICS\LAB\Mall_Customers.csv')
df
```

```
Out[5]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows x 5 columns

```
In [6]: # INDEPENDENT VARIABLE
x = df.iloc[:,3:5].values
x
```

```
Out[6]: array([[ 15,  39],
[ 15,  81],
[ 16,   6],
[ 16,  77],
[ 17,  40],
[ 17,  76],
[ 18,   6],
[ 18,  94],
[ 19,   3],
[ 19,  72],
[ 19,  14],
[ 19,  99],
[ 20,  15],
[ 20,  77],
[ 20,  13],
[ 20,  79],
[ 21,  35],
[ 21,  66],
[ 23,  29],
```

```
[ 23,  98],
[ 24,  35],
[ 24,  73],
[ 25,   5],
[ 25,  73],
[ 28,  14],
[ 28,  82],
[ 28,  32],
[ 28,  61],
[ 29,  31],
[ 29,  87],
[ 30,   4],
[ 30,  73],
[ 33,   4],
[ 33,  92],
[ 33,  14],
[ 33,  81],
[ 34,  17],
[ 34,  73],
```

```
[ 37,  26],
[ 37,  75],
[ 38,  35],
[ 38,  92],
[ 39,  36],
[ 39,  61],
[ 39,  28],
[ 39,  65],
[ 40,  55],
[ 40,  47],
[ 40,  42],
[ 40,  42],
[ 42,  52],
[ 42,  60],
[ 43,  54],
[ 43,  60],
[ 43,  45],
[ 43,  41],
[ 44,  50],
[ 44,  46],
```

```
[ 44,  50],
[ 44,  46],
[ 46,  51],
[ 46,  46],
[ 46,  56],
[ 46,  55],
[ 47,  52],
[ 47,  59],
[ 48,  51],
[ 48,  59],
[ 48,  50],
[ 48,  48],
[ 48,  59],
[ 48,  47],
[ 49,  55],
[ 49,  42],
[ 50,  49],
[ 50,  56],
[ 54,  47],
[ 54,  54],
```


[54, 53],	[60, 47],	[65, 50],	[73, 5],	[78, 20],
[54, 48],	[60, 50],	[65, 43],	[73, 88],	[78, 76],
[54, 52],	[61, 42],	[65, 59],	[73, 7],	[78, 16],
[54, 42],	[61, 49],	[67, 43],	[73, 73],	[78, 89],
[54, 51],	[62, 41],	[67, 57],	[74, 10],	[78, 1],
[54, 55],	[62, 48],	[67, 56],	[74, 72],	[78, 78],
[54, 41],	[62, 59],	[67, 40],	[75, 5],	[78, 1],
[54, 44],	[62, 55],	[69, 58],	[75, 93],	[78, 73],
[54, 57],	[62, 56],	[69, 91],	[76, 40],	[79, 35],
[54, 46],	[62, 42],	[70, 29],	[76, 87],	[79, 83],
[57, 58],	[63, 50],	[70, 77],	[77, 12],	[81, 5],
[57, 55],	[63, 46],	[71, 35],	[77, 97],	[81, 93],
[58, 60],	[63, 43],	[71, 95],	[77, 36],	[85, 26],
[58, 46],	[63, 48],	[71, 11],	[77, 74],	[85, 75],
[59, 55],	[63, 52],	[71, 75],	[78, 22],	[86, 20],
[59, 41],	[63, 54],	[71, 9],	[78, 90],	[86, 95],
[60, 49],	[64, 42],	[71, 75],	[78, 17],	[87, 27],
[60, 40],	[64, 46],	[72, 34],	[78, 88],	[87, 63],
[60, 42],	[65, 48],	[72, 71],	[78, 20],	[87, 13],
[60, 52],	[65, 50],	[73, 5],		[87, 75],

```
In [10]: a=[]
for i in range(1,11):
    kmeans = KMeans(n_clusters=i,init='k-means++')
    kmeans.fit(x)
    a.append(kmeans.inertia_)
```

C:\Users\Lenovo\anaconda3\lib\site-packages\sklearn\cluster_kmeans.py:881: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(

```
In [11]: #euclidean distance
a
```

```
Out[11]: [269981.280000000014,
182440.30762987016,
106348.37306211119,
73679.78903948837,
44448.45544793369,
37239.83554245604,
30241.34361793659,
24986.52564064289,
21850.16528258562,
20087.616325699386]
```

No of clusters	Euclidean distance
1	260000
2	180000
3	105000
4	75000
5	45000
6	35000
7	30000
8	25000
9	22000
10	20000

[illegible]

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	
0	1	Male	19	15	39	4
1	2	Male	21	15	81	0
2	3	Female	20	16	6	4
3	4	Female	23	16	77	0
4	5	Female	31	17	40	4
...	...	...	...	...	...	...
195	198	Female	35	120	79	2
196	197	Female	45	126	26	3
197	198	Male	32	126	74	2
198	199	Male	32	137	16	3
199	200	Male	30	137	83	2

A scatter plot showing the relationship between Speed (Y-axis, 0 to 100) and Salary (X-axis, 20 to 140). The data points are colored by car model: red (Ferrari), green (Lexus), blue (Porsche), yellow (Toyota), and black (VW). The plot shows that higher speed generally correlates with higher salary, with the Ferrari (red) having the highest speed and salary, and the VW (black) having the lowest speed and salary.

Practical-6

Aim: Perform the Decision tree classification algorithm using a dataset.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.preprocessing import LabelEncoder

# Load the CSV file into a DataFrame
csv_file_path = 'path_to_your_file/marksheet.csv' # Update with your file path
df = pd.read_csv(csv_file_path)

# Preprocess the data
# Assume 'Section' is the target variable for classification
target = 'Section'
features = df.drop(columns=[target])

# Encode categorical variables
label_encoder = LabelEncoder()
df['Gender'] = label_encoder.fit_transform(df['Gender'])
df[target] = label_encoder.fit_transform(df[target])

# Define features and target
X = df.drop(columns=[target])
y = df[target]

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Train the Decision Tree model
clf = DecisionTreeClassifier(random_state=42)
clf.fit(X_train, y_train)

# Make predictions on the test set
y_pred = clf.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
conf_matrix = confusion_matrix(y_test, y_pred)
class_report = classification_report(y_test, y_pred)

# Print the results
print(f'Accuracy: {accuracy}')
print("Confusion Matrix:\n", conf_matrix)
print("Classification Report:\n", class_report)
```

Enrollment No:

Practical-7

Aim: Study and installation of the tools like PowerBI tool for data Visualization.

Introduction to Microsoft Power BI

There are several tools that make it easier for us to analyze the trajectory of business assets using data. However, Microsoft Power BI is something so unique that makes it super easy to make data-driven decisions.

It is one of the oldest, most famous, and most accessible tools for Data Visualization and business intelligence. From different sources, it collects and helps to convert into an interactive dashboard and BI reports. It is available as a Power BI Desktop, and mobile but also available for it service based on Saas. Business users use a set of services provided by it to create reports and achieve a goal. It apps help in generating reports, whereas its services help in publishing reports. It is a cloud-based service provided by Microsoft that helps viewers to visualize data, and this process is called Data visualization with it. It helps in sorting, analyzing, and comparing data more frequently and quickly. It is compatible with various resources like Microsoft Excel, Access Database, SQL Server, which helps make the best choice for data analysts.

Installation process of Power BI

Two Ways to install Power BI and Get it up and Running:

Either of the two approaches gets the latest version of Power BI Desktop onto your computer.

1. Install as an app from the Microsoft Store.

There are a few ways to access the most recent version of Power BI Desktop from the Microsoft Store. Use one of the following options to open the Power BI Desktop page of the Microsoft Store:

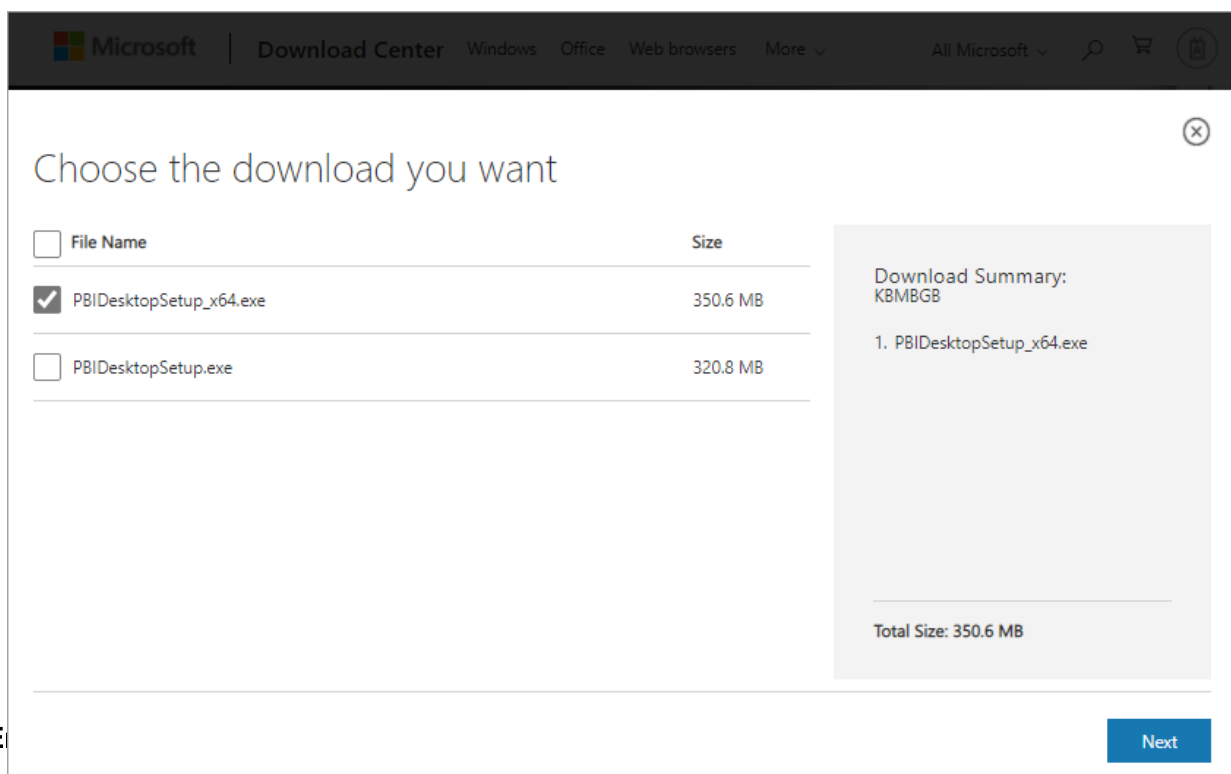
1. Open a browser and go directly to the [Power BI Desktop page](#) of the Microsoft Store.
2. From the [Power BI service](#), select the Download icon in the upper right corner and then choose Power BI Desktop.
3. Go to the [Power BI Desktop product page](#) and select Download Free.

4. After you've landed on the Power BI Desktop page of the Microsoft Store, select Install.



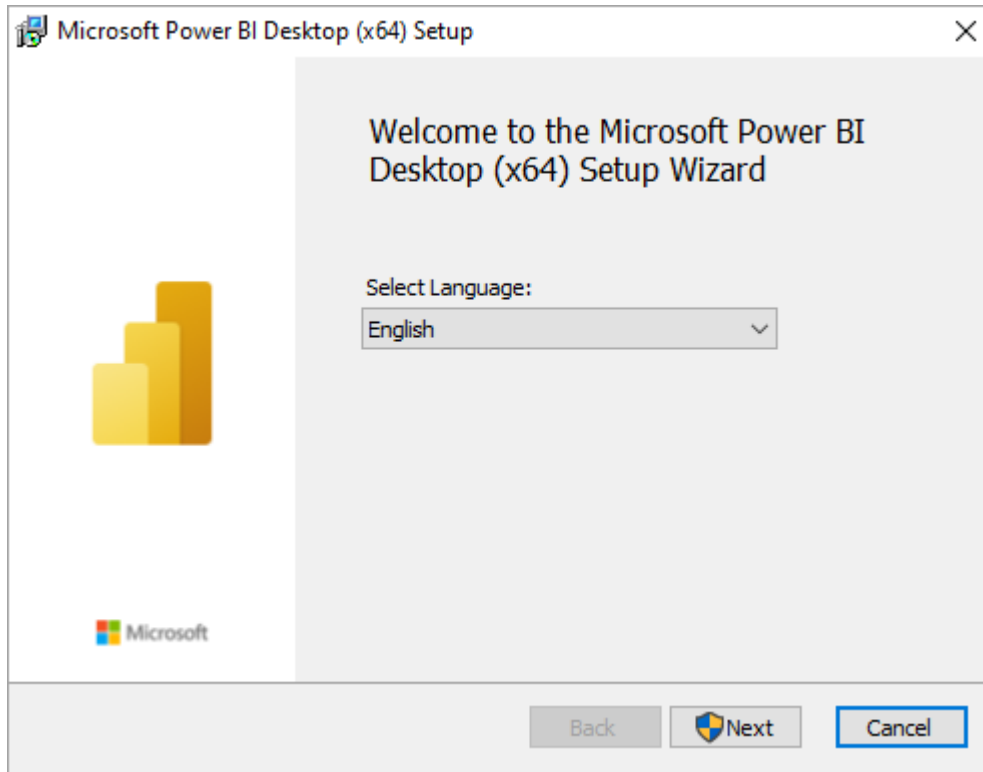
2. **Download directly as an executable; you download and install it on your computer.**

Select Download from the Download Center to download the Power BI Desktop executable from the [Download Center page](#). Then, specify a 32-bit or 64-bit installation file to download.



Install Power BI Desktop after downloading it

You're prompted to run the installation file after downloading it.



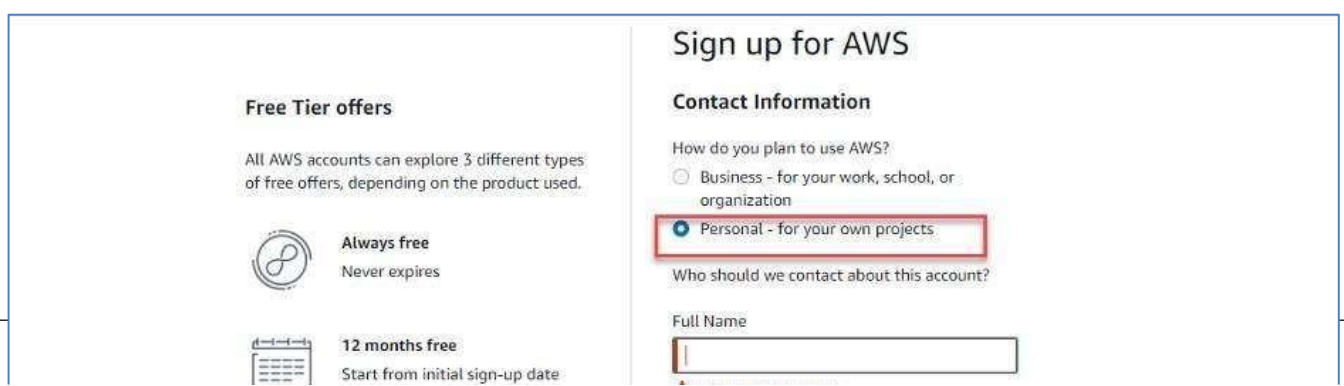
And that's it; you are good to go.

Note: Windows Users can ignore the instructions beyond this point. The instructions ahead are specifically for MAC users.

Installing Instructions for MAC

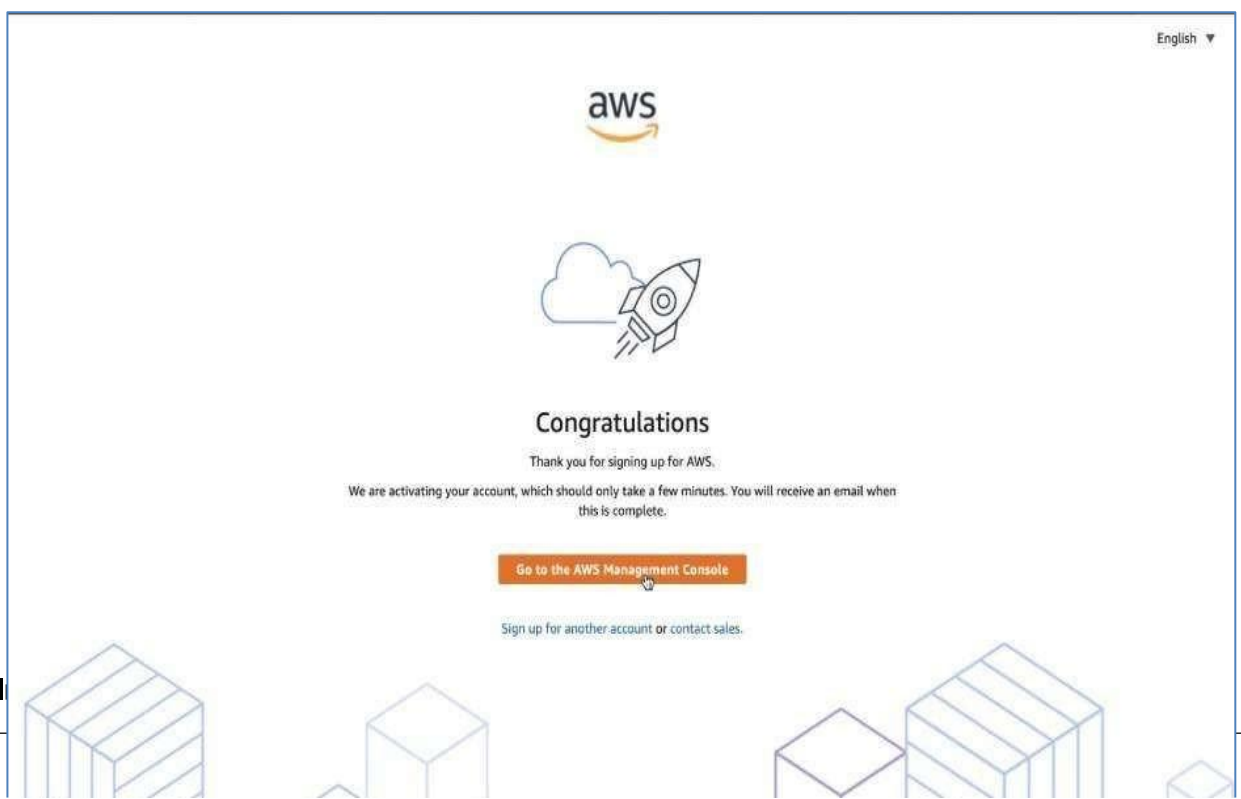
Since Power BI is not directly available for mac users, we will use an AWS container to run an instance of windows ten where you can download power BI and install it for use.

1. Create an AWS account using this URL <https://portal.aws.amazon.com/billing/signup#/start>
2. Select Personal when asked how you plan to use this account



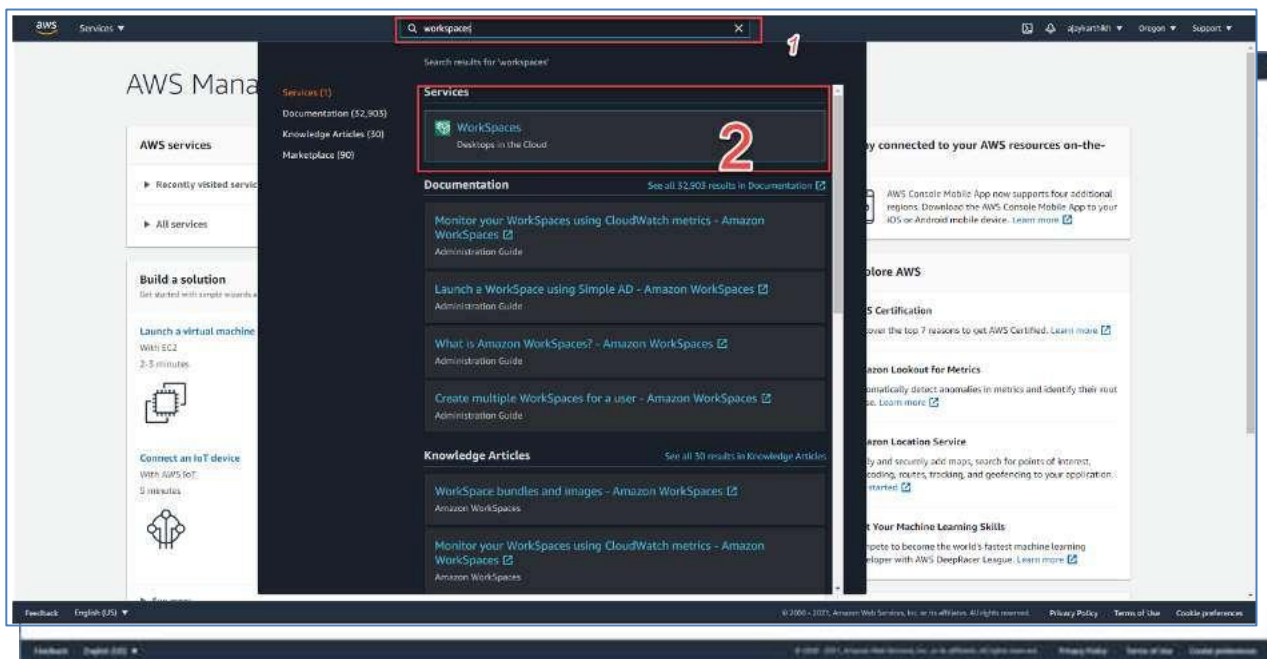
3. After completing all the steps, you will be asked to enter billing information, which is a mandatory step.

Enroll

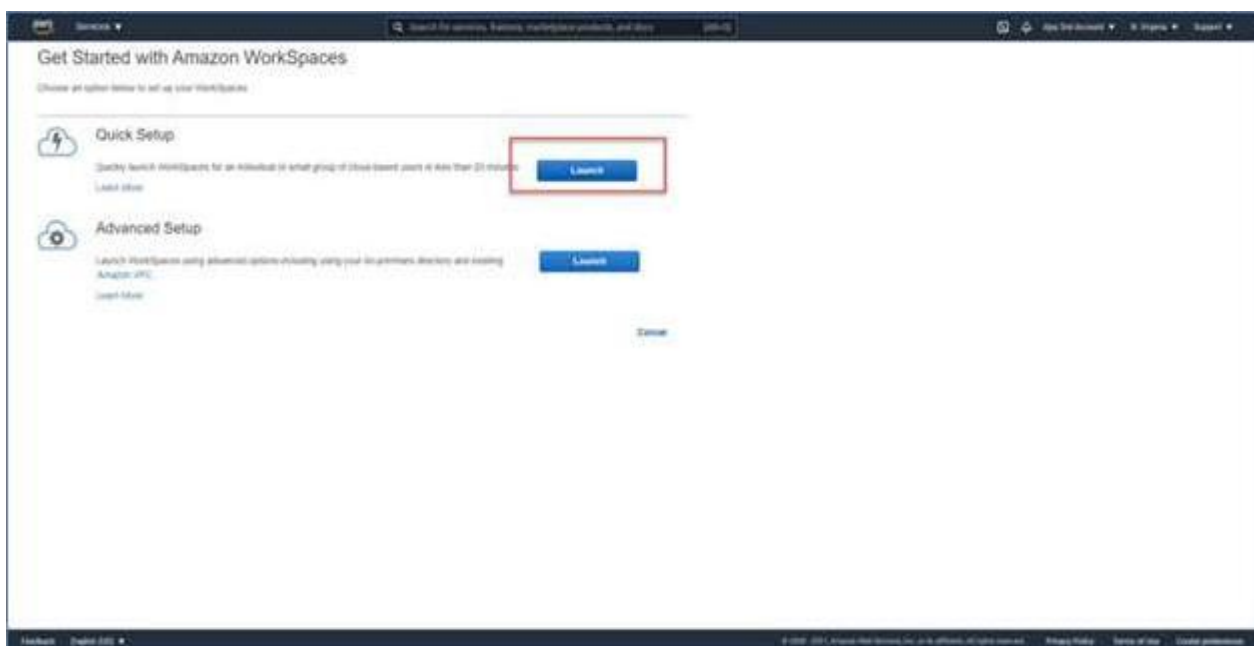
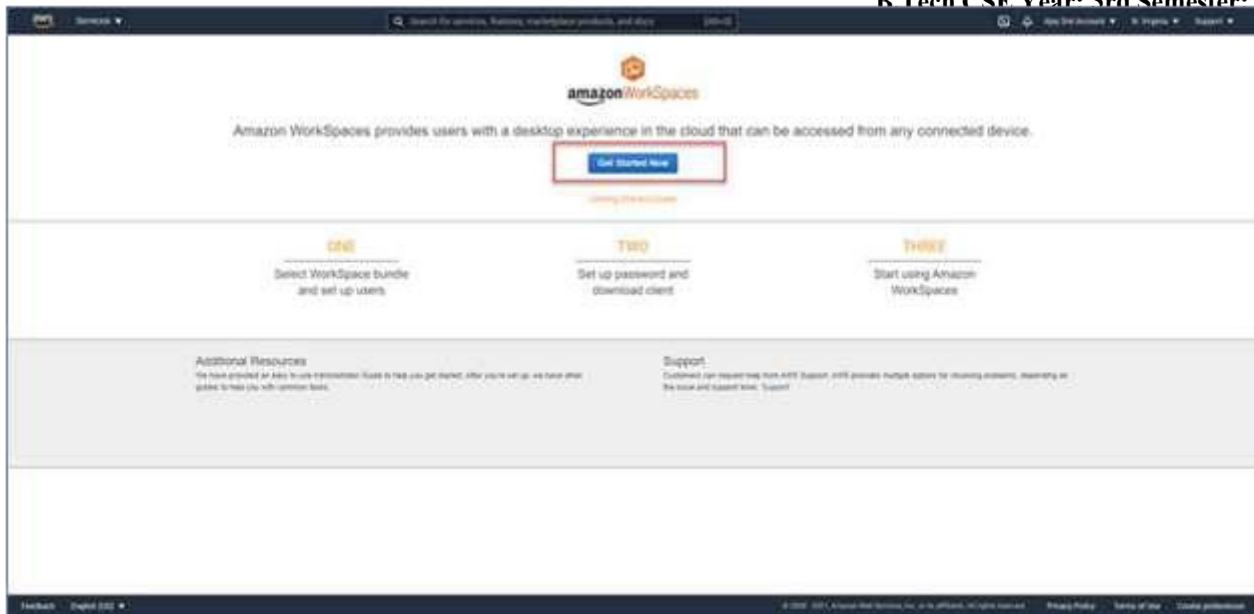


4. Login into your AWS console, search for “workspaces,” and click on the Workspaces icon as shown below.

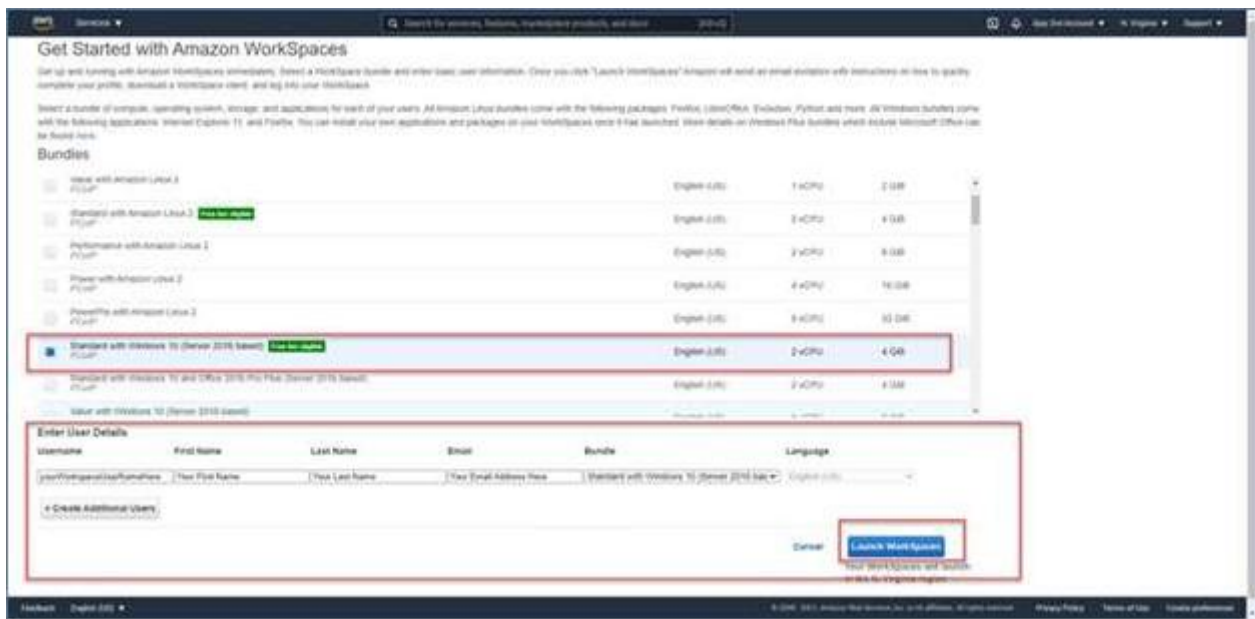
5. Here you should be able to see a Launch Workspace.



Enrollment No:



Enrollment No:

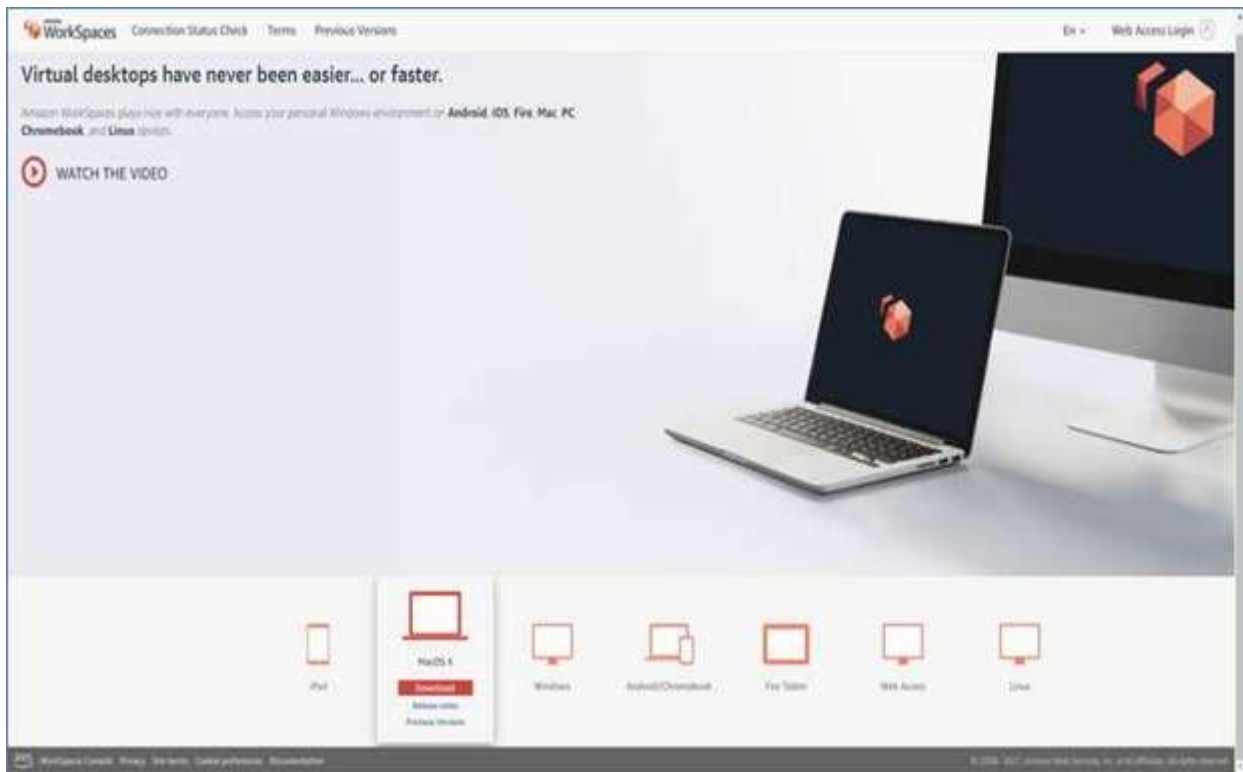


When you click launch, you will be prompted to view the workspace console; clicking on this button will take you to the console where it says that it will take 20 min to set the workspace; typically, it takes 15-30 min, my experience.

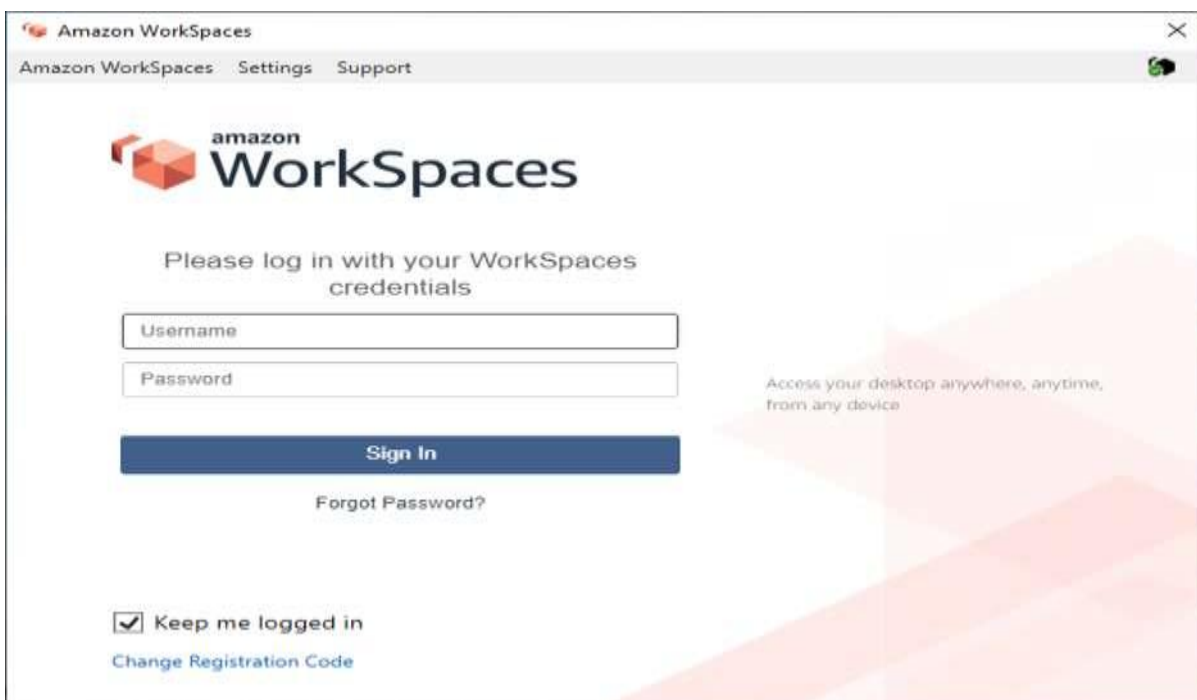


14. Now in this waiting time, download the appropriate Workspace RD client from this link - https://clients.amazonworkspaces.com/?refid=ps_a134p000006vmxaaai&trkcampain=acq_paid_search_brand

Enrollment No:

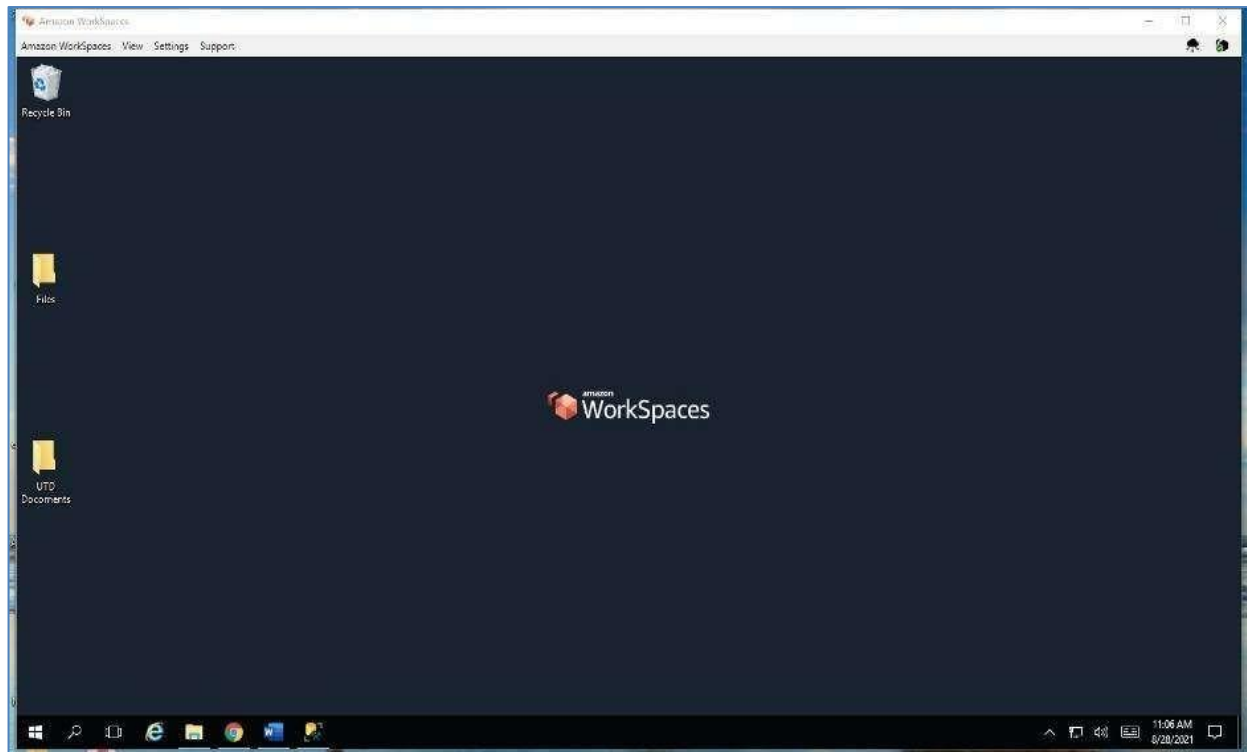


15. Now Install this and wait for your Workspace to finish setting up
16. Start the Amazon Workspace client on your Mac, and you should see something like the below screenshot



Enrollment No:

14. Remember that you did not create a password in step 11. It would help if you were getting an email to create a password. Use that password and the username used in step 11 here.
15. After successful login, you should see something like this.



20. Select Download from the Download Center to download the Power BI Desktop executable from the Download Center page. Then, specify a 32-bit or 64-bit installation file to download.
21. Follow all the steps followed for Power BI installation on Windows.

Practical-8

Aim: Load a dataset from different sources in PowerBI and apply transformations to it.

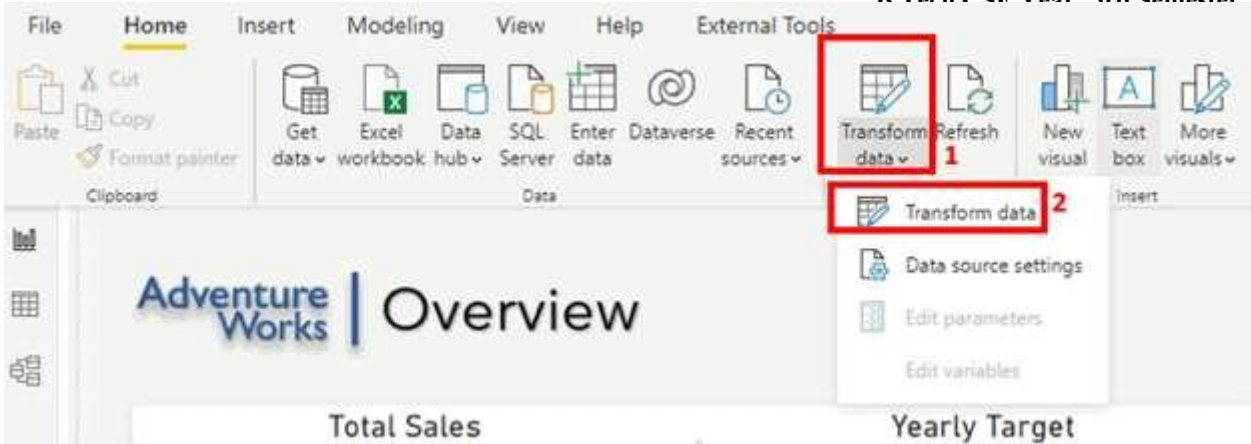
Step 1: Open the Power BI report in Power BI Desktop.

First, you need to open the Power BI report that you want to update in Power BI Desktop.

Step 2: Click **Transform data**.

Navigate to the “Data” tab, click the **Transform data** drop-down, and then **Transform data** to get to Power Query.

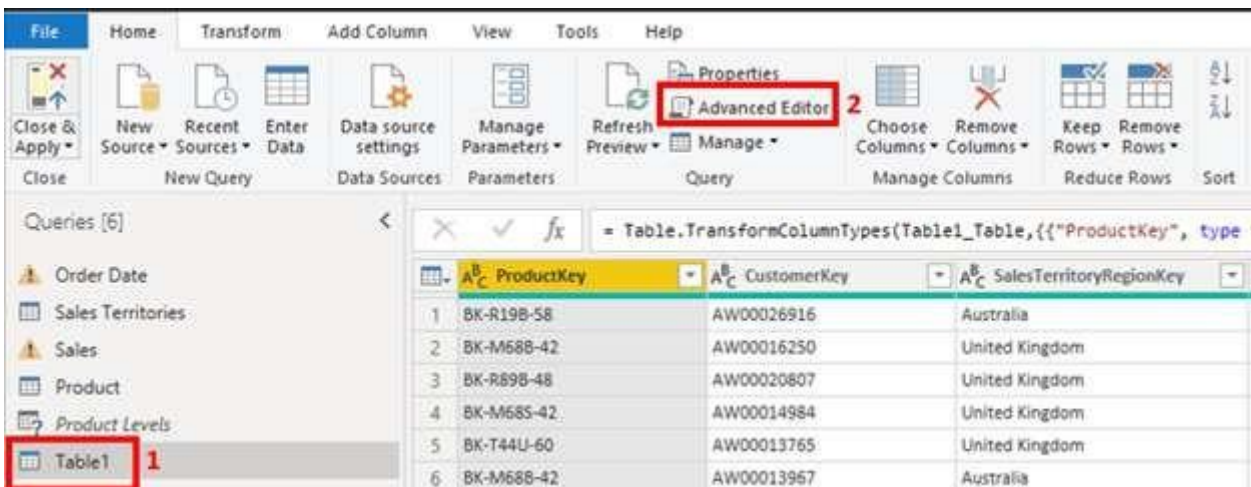
Enrollment No:



Step 3: Click on **New Source** and connect to the new data source. Confirm the column names in Table 1 (new data source) are the same as that of the Sales table (current data source).

Step 4: Click on Advanced Editor.

Select **Table 1**, navigate to the “View” tab, and click on **Advanced Editor**.

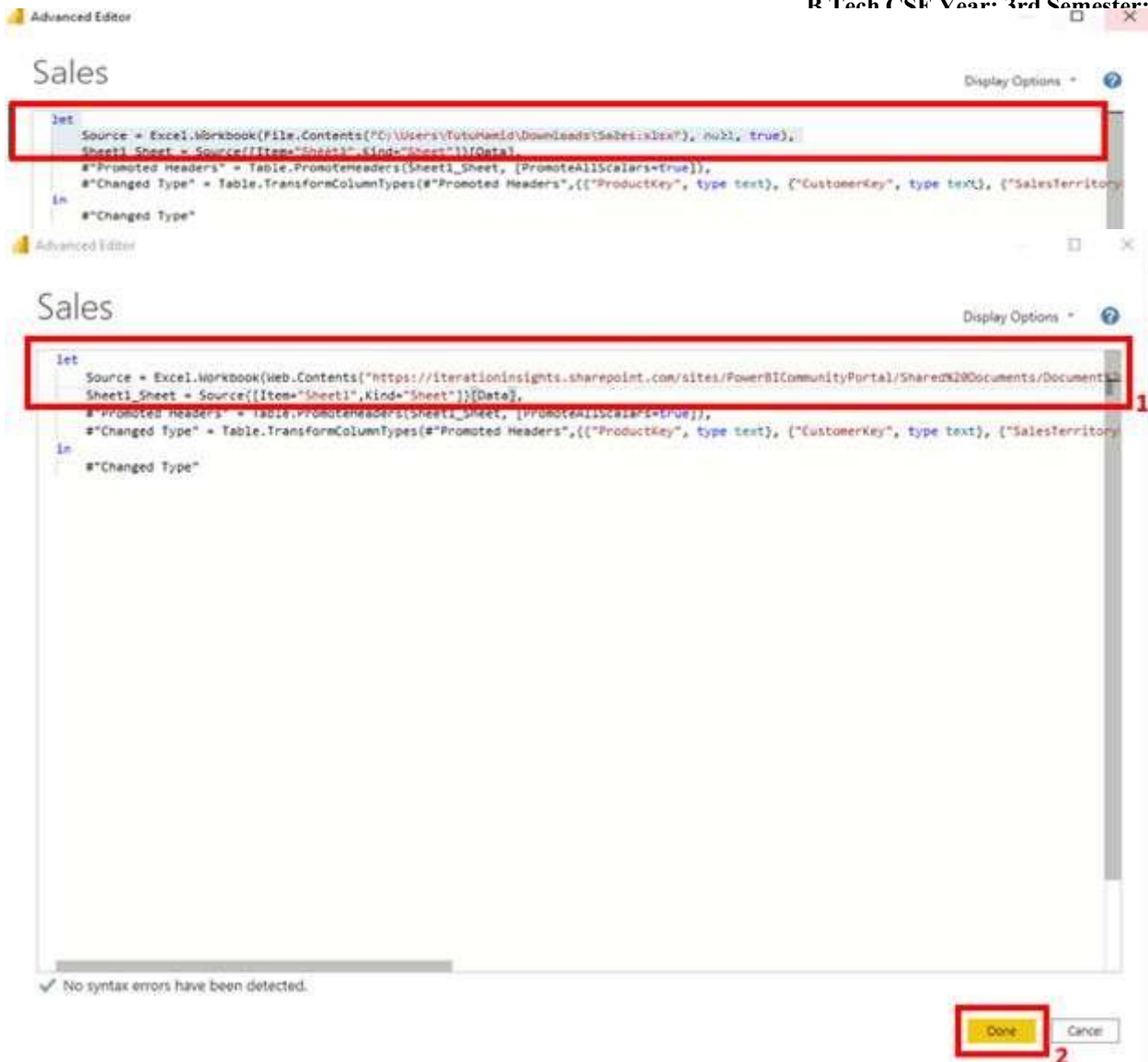


Step 5: Copy the M code up to [data].

Locate the M code that specifies the new data source and copy from beginning up to [data].



Step 6: Select Sales table and click Advance Editor. Replace the M code up to [data] with the code you copied from Table1 and click done.



After making the necessary changes, the user should review the M code and ensure it accurately reflects the new data source.

Step 7: Delete Table1 and click **Close & Apply**.

Practical-9

Aim: Study and Plot various graphs for Data Visualization on PowerBI.

Scatter charts

Scatter charts work well in many scenarios:

- Show relationships between two numerical values.

Enrollment No:

- Plot two groups of numbers as one series of x and y coordinates.
- Display worksheet data with pairs or grouped sets of values.
- Show patterns in large sets of data.
- Compare large amounts of data points irrespective of time measurements.
- Convert horizontal axis into logarithmic scale.
- Substitute for line charts to enable changing horizontal axis scale.

Bubble charts

You can use a bubble chart in many of the same scenarios as a scatter chart. Here are some of the other ways you can use bubble charts:

- Visually emphasize value differences with variable bubble size.
- Support scenarios with three data series that each has sets of values.
- Present financial data in a visual rather than numerical form.
- Display data with quadrants.

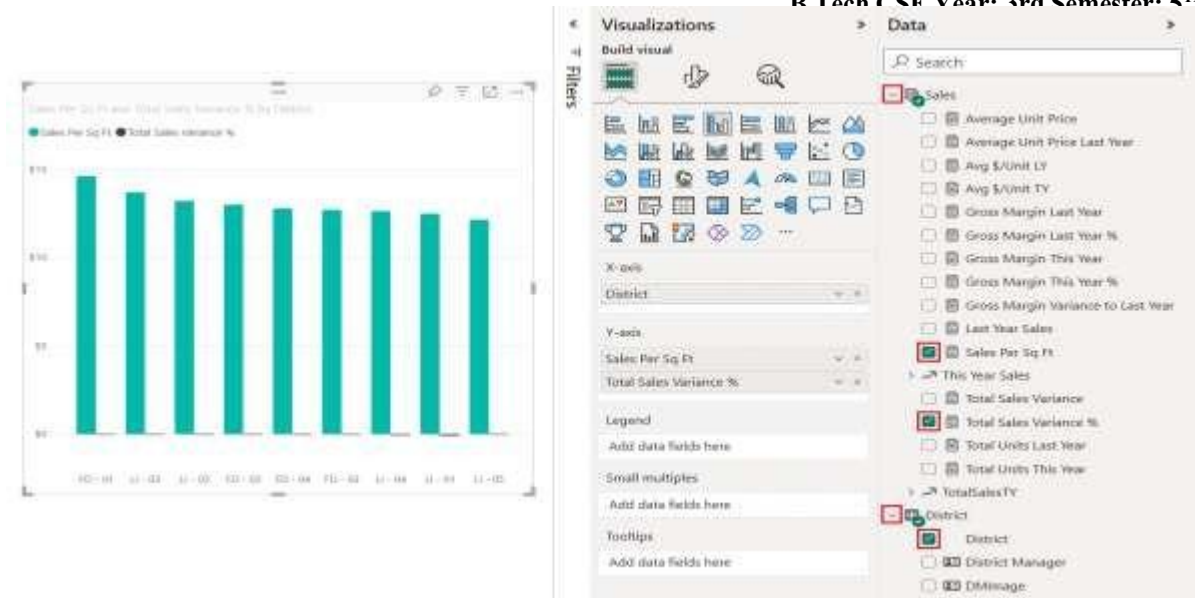
Dot plot charts

Use cases for the dot plot chart are similar to the scenarios described for scatter and bubble charts. The primary advantage of dot plot charts is the ability to include categorical data along the horizontal axis.

Create a scatter chart

1. On the **Data** pane, select three fields:
 - Expand **Sales** and select the **Sales Per Sq Ft** and **Total Sales Variance %** checkboxes.
 - Expand **District** and select the **District** checkbox.

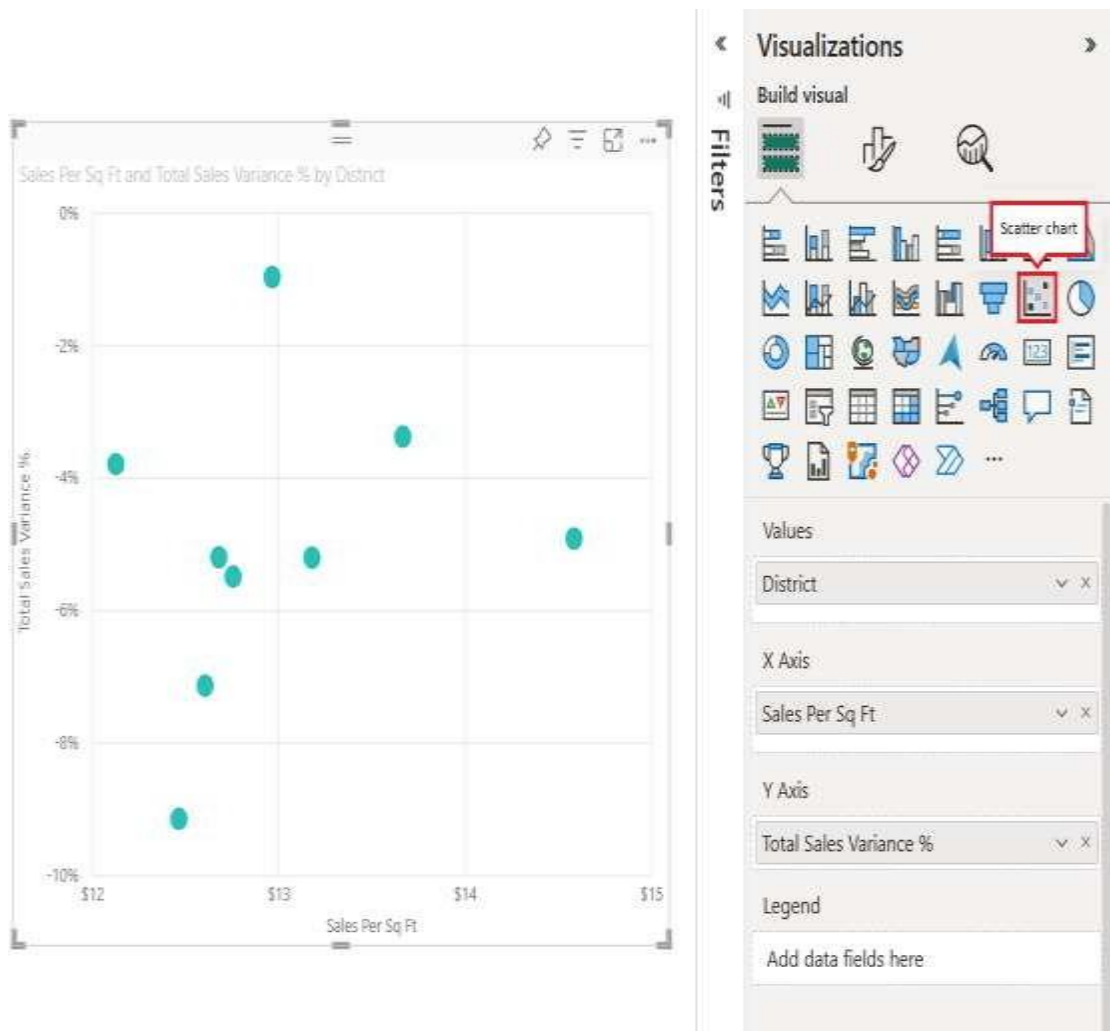
By default, Power BI creates a clustered column chart to display the data. On the **Visualizations** pane, the **District** field is assigned to the **X-axis** and the other two fields are assigned to the **Y-axis**.



You can now convert the clustered column chart visual into a scatter chart.

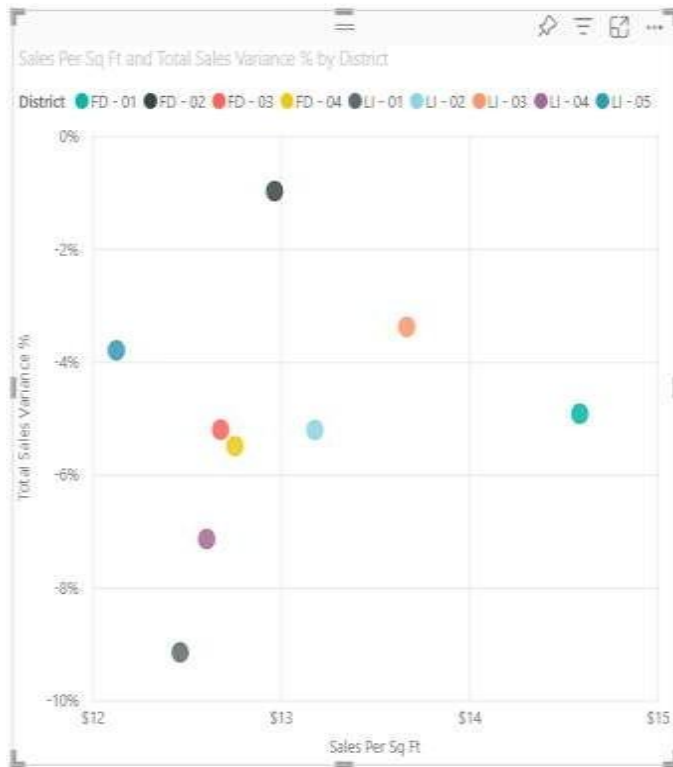
2. Select the chart visual, and then select **Scatter chart** on the **Visualizations** pane.

Notice the changes to the **Visualizations** pane. The **District** field is now listed under **Values**. The chart axes are also different. Make sure that Power BI plots the **Sales Per Sq Ft** field along the **X Axis** and the **Total Sales Variance %** field along the **Y Axis**.



3. On the **Visualizations** pane, drag the **District** field from the **Values** section to the **Legend** section.

Power BI creates data points where the data values intersect along the x and y axes. The data point colors represent different districts.



'District' [District]

Visualizations

Build visual

Filters

Values

Add data fields here

X Axis

Sales Per Sq Ft

Y Axis

Total Sales Variance %

Legend

District

Size

Practical-10

Aim: Given a case study: Interactive Data Analytics with Power BI Dashboard

1. HEATHROW

- ☐ Heathrow airport is an international airport in London. It is the second busiest international airport in the world after Dubai international airport. And, also the seventh-largest in terms of total passenger traffic.



THE CHALLENGE

- ☐ Being the world's seventh busiest airport in overall passenger traffic, one can only imagine the level of efficiency and efforts expected from the airport's ground management to keep the airport functioning properly. Managing over 2,00,000 passengers every day can be quite a challenging task for airport authorities and ground staff. Every department needs to be in absolute coordination and sync to be able to manage the passenger traffic and give them a smooth experience at the airport. At such busy airports, every day brings new challenges and uncertainties with it. Unexpected disruptions in the smooth workflow of operations at the airport disturb the entire functioning. Issues can arise due to stormy weather, delayed flights, canceled flights, shifts in jet streams, etc. disturbing the airport's smooth functioning. Such problems send the passengers as well as airport employees into turmoil.
- ☐ The airport needed a central digitalized management system as a solution to this problem. Such a system would use the large amounts of data being produced by operational systems at the airport and transform it into useful visual insights. The interpretations produced by the BI tool can be used by airport staff for better functioning and passenger management.

THE CHANGE

- ☐ Heathrow group went with Microsoft Power BI as their business intelligence software and Microsoft Azure for cloud services. The airport has deployed Microsoft Azure technology to collect data from back-end operational systems at the airport. These systems are *check-in counters, baggage tracking systems, flight schedules, and weather tracking systems, cargo tracking* and many more.
- ☐ The operational data from these systems are forwarded to business intelligence platforms like Power BI. In Power BI, users shape this data into useful information that the airport staff can use.

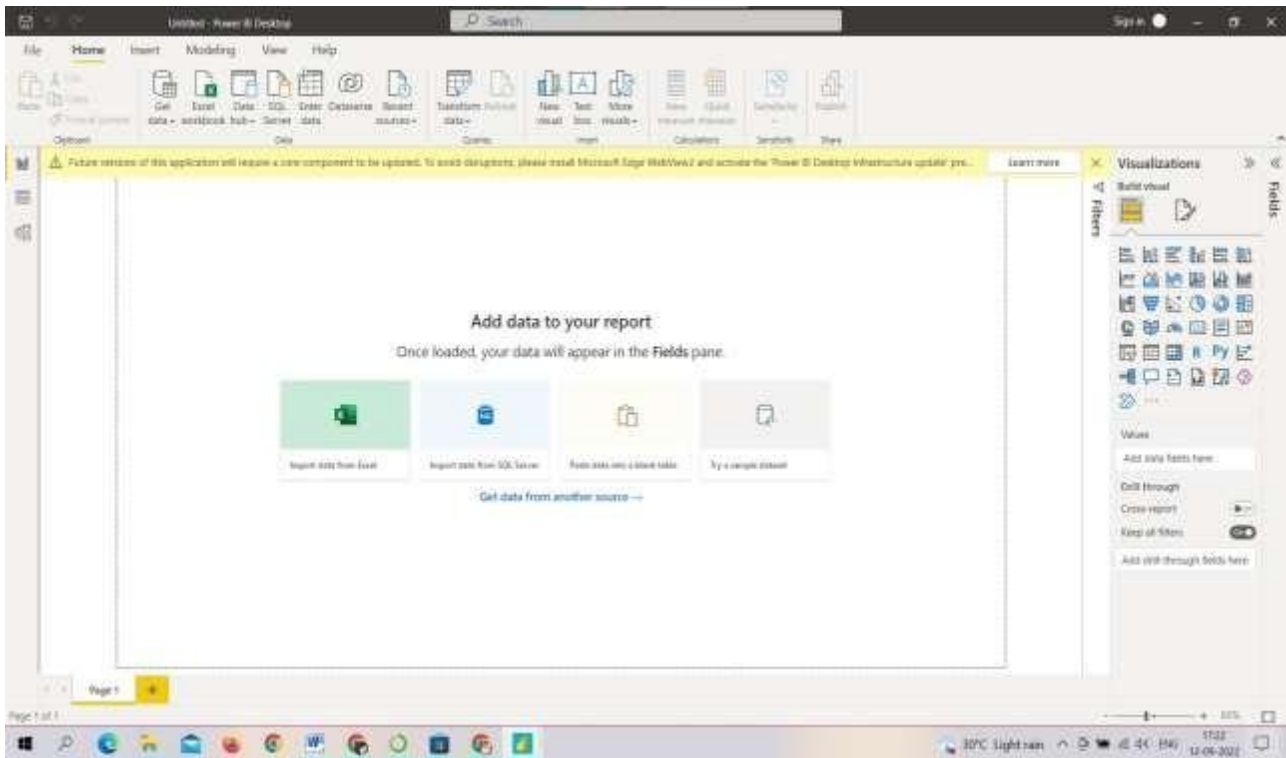
- ☐ Power BI transforms the crude information into informative visuals showing different statuses and statistics of the airport systems. Then, the ground staff like baggage handlers, gate agents, air traffic controllers, etc. use this information to properly operate and manage passengers.
- ☐ Services such as Azure Stream Analytics, Azure Data Lake Analytics, and Azure SQL Database are used to *extract, clean and prepare operational data* in real-time. This data is about *flight movements, security queues, passenger transfers, and immigration queues*. Ultimately, Power BI uses data from these Azure services for analysis and interpretation.
- ☐ Operational data from different data sources come into Power BI. Then Power BI tools are used to transform that data into meaningful insights with the help of visual reports, graphics, and dashboards. About 75,000 airport employees have information on their fingertips by the virtue of Power BI.
- ☐ Let us understand this with the help of a **real-world example**. If there is a change in the jet stream, it may delay about 20 flights in a day. This will result in about 6,000 passengers waiting at the airport at a given point of time. It will increase passenger traffic and density at the airport. Power BI works like the centralized information system. The airport uses it to inform about the sudden passenger influx. This information goes out to different sections such as *food outlets, immigration, customs, gate attenders, baggage handlers* at the airport. This will give them time to prepare themselves to attend the passengers.
- ☐ With the presence of smart BI solutions like Power BI, airport staff is notified in advance about the probable delays and the sudden rush of passengers at the airport. This help management groups and other employees to take suitable actions in advance like *increasing the food stock, adding extra passenger buses, increasing the ground staff, directing the passengers to the waiting area*, etc. to avoid any last-minute hustle.
- ☐ Thus, with the help of a powerful BI tool like Power BI, Heathrow has been benefited in more than one way. They are extremely happy and satisfied with the capabilities of Power BI helping them give a hassle-free airport experience to their passengers. Heathrow also is extending Power BI applications by trying to anticipate passenger flow at the airport to avoid any unexpected disruptions for the passengers.

STEPS TO ANALYZE DATA IN POWER BI

Step1: Download the Power BI Software in your PC

Step2: After that download the dataset for performing data analysis. Here I have downloaded Sales of chocolates to importing to different countries and containing info about the dealer

Step 3: Then open Power BI. The home page will look like below page



Step 4: Then import file which you want

Step 5: Then according to you do data analysis

